Lecture Notes:
Induction and Recursion

Discrete Mathematics

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Section 1 Mathematical Induction

1.1 Principle of Mathematical Induction

Theorem (Principle of Mathematical Induction). Let $P(n)$ be a propositional function defined for integers $n \geq n_0$, where n_0 is a fixed integer (usually $n_0 = 0$ or 1). Suppose that:

(i) **Base case:** $P(n_0)$ is true.

(ii) **Inductive step:** For every integer $k \geq n_0$, if $P(k)$ is true, then $P(k+1)$ is true.

Then $P(n)$ is true for all integers $n \geq n_0$.

Remark. Mathematical induction is a fundamental proof technique for statements involving natural numbers. It formalizes the "domino effect": if the first domino falls, and each domino knocks over the next one, then all dominoes will eventually fall.

1.2 Structure of an Inductive Proof

A proof by mathematical induction consists of the following essential components:

1. **Define the predicate:** Clearly state $P(n)$, the statement to be proved for each $n \geq n_0$.
2. **Base case:** Verify that $P(n_0)$ holds. This anchors the induction.
3. **Inductive hypothesis:** Assume that $P(k)$ is true for some arbitrary $k \geq n_0$.
4. **Inductive step:** Using the inductive hypothesis, prove that $P(k+1)$ is true.
5. **Conclusion:** By the principle of mathematical induction, $P(n)$ holds for all $n \geq n_0$.

1.3 Examples of Proofs by Induction

Prove that for all $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the statement: $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Base case ($n = 1$): Left side: 1. Right side: $\frac{1 \cdot 2}{2} = 1$. So $P(1)$ is true.
Inductive hypothesis: Assume $P(k)$ is true for some $k \geq 1$, i.e.,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Inductive step: We must prove $P(k+1)$:

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Starting from the left side and using the inductive hypothesis:

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= (k+1)\left(\frac{k}{2} + 1\right) \\ &= (k+1)\left(\frac{k+2}{2}\right) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus $P(k+1)$ holds.

Conclusion: By mathematical induction, $P(n)$ is true for all $n \geq 1$.

Prove that $n! > 2^n$ for all integers $n \geq 4$.

Proof. Let $P(n)$ be: $n! > 2^n$.

Base case ($n = 4$): $4! = 24$, $2^4 = 16$, and $24 > 16$. So $P(4)$ holds.

Inductive hypothesis: Assume $P(k)$ for some $k \geq 4$, i.e., $k! > 2^k$.

Inductive step: Show $(k+1)! > 2^{k+1}$.

$$\begin{aligned}
(k+1)! &= (k+1) \cdot k! \\
&> (k+1) \cdot 2^k \quad \text{(by inductive hypothesis)} \\
&\geq 5 \cdot 2^k \quad \text{(since } k \geq 4,\ k+1 \geq 5\text{)} \\
&> 2 \cdot 2^k = 2^{k+1}.
\end{aligned}$$

Thus $P(k+1)$ holds. By induction, $n! > 2^n$ for all $n \geq 4$.

1.4 Strong Induction

Theorem (Principle of Strong Induction). Let $P(n)$ be a propositional function for $n \geq n_0$. Suppose that:

(i) **Base case:** $P(n_0)$ is true.

(ii) **Strong inductive step:** For every $k \geq n_0$, if $P(n_0), P(n_0+1), \ldots, P(k)$ are all true, then $P(k+1)$ is true.

Then $P(n)$ is true for all $n \geq n_0$.

Remark. Strong induction differs from ordinary induction in the hypothesis: we assume the truth of P for *all* previous values, not just the immediate predecessor. This is especially useful when $P(k+1)$ depends on multiple earlier values, not just $P(k)$.

Prove that every integer $n \geq 2$ is either prime or can be written as a product of primes.

Proof (by strong induction). *Base case ($n = 2$):* 2 is prime, so the statement holds.

Inductive hypothesis: Assume the statement holds for all integers m with $2 \leq m \leq k$ for some $k \geq 2$.

Inductive step: Consider $k+1$. If $k+1$ is prime, we are done. If not, then $k+1 = ab$ where $2 \leq a, b \leq k$. By the inductive hypothesis, both a and b are either prime or products of primes. Hence $k+1$ is a product of primes. This completes the proof.

Section 2 Recursively Defined Sequences

Definition (Recursion). A sequence $\{a_n\}$ is said to be *recursively defined* (or defined by *recurrence*) if each term is defined in terms of previous terms. A recursive definition consists of two parts:

(i) **Initial condition(s):** The value(s) of the first one or more terms, given explicitly.

(ii) **Recurrence relation:** An equation that expresses a_n in terms of earlier terms $a_{n-1}, a_{n-2}, \ldots$ for all n beyond the initial conditions.

Remark. Recursive definitions are ubiquitous in mathematics and computer science. They allow complex objects to be built from simpler ones, and they naturally lead to inductive proofs of properties.

[Factorial] The factorial function $n!$ can be defined recursively:

$$0! = 1 \quad \text{(initial condition)},$$
$$n! = n \cdot (n-1)! \quad \text{for } n \geq 1 \quad \text{(recurrence relation)}.$$

[Arithmetic Progression] An arithmetic progression with first term a and common difference d:

$$a_0 = a,$$
$$a_n = a_{n-1} + d \quad \text{for } n \geq 1.$$

[Geometric Progression] A geometric progression with first term a and common ratio r:

$$a_0 = a,$$
$$a_n = r \cdot a_{n-1} \quad \text{for } n \geq 1.$$

[Fibonacci Sequence] The famous Fibonacci sequence $\{F_n\}_{n=0}^{\infty}$ is defined recursively:

$$F_0 = 0, \quad F_1 = 1 \quad \text{(initial conditions)},$$
$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2 \quad \text{(recurrence relation)}.$$

The first few terms are: $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

Interpretation: Each Fibonacci number (after the first two) is the sum of the two preceding numbers. The sequence appears in numerous natural phenomena, algorithm analysis, and combinatorial problems.

Section 3 Recurrence Relations

Definition (Recurrence Relation). A *recurrence relation* for a sequence $\{a_n\}$ is an equation that expresses a_n as a function of one or more of the preceding terms $a_{n-1}, a_{n-2}, \ldots, a_{n-k}$ for all $n \geq k$, where k is a fixed integer. The relation must hold for all n beyond some initial index.

Definition (Order of a Recurrence). The *order* of a recurrence relation is the number of preceding terms on which a_n depends. For example, $a_n = a_{n-1} + a_{n-2}$ has order 2.

3.1 Types of Recurrence Relations

(1) **Linear vs. Nonlinear:** A recurrence is *linear* if it is a linear combination of previous terms. Otherwise, it is nonlinear.

Example: $a_n = 3a_{n-1} - 2a_{n-2}$ is linear. $a_n = a_{n-1}^2 + a_{n-2}$ is nonlinear.

(2) **Homogeneous vs. Non-homogeneous:** A linear recurrence is *homogeneous* if the linear combination of previous terms equals zero when we bring all terms involving a to one side. It is *non-homogeneous* if there is an additional term that is a function of n alone.

Example: $a_n = 5a_{n-1} - 6a_{n-2}$ is homogeneous. $a_n = 5a_{n-1} - 6a_{n-2} + 2^n$ is non-homogeneous.

(3) **Constant vs. Variable Coefficients:** If the coefficients multiplying the previous terms are constants (independent of n), we say the recurrence has *constant coefficients*. If they depend on n, they are *variable coefficients*.

Example: $a_n = 3a_{n-1} - 2a_{n-2}$ has constant coefficients. $a_n = na_{n-1} + a_{n-2}$ has variable coefficients.

Definition (Linear Homogeneous Recurrence with Constant Coefficients). A *linear homogeneous recurrence relation with constant coefficients* of order k has the form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k},$$

where $c_1, c_2, \ldots, c_k \in \mathbb{R}$ are constants and $c_k \neq 0$. This is the main class of recurrence relations for which systematic solution methods exist.

The Fibonacci recurrence $F_n = F_{n-1} + F_{n-2}$ is a linear homogeneous recurrence with constant coefficients of order 2 (here $c_1 = 1$, $c_2 = 1$).

The recurrence $a_n = 2a_{n-1} - a_{n-2}$ is a linear homogeneous recurrence with constant coefficients of order 2.

[Non-homogeneous] $a_n = a_{n-1} + a_{n-2} + n$ is linear but non-homogeneous due to the $+n$ term.

Section 4 Solving Recurrence Relations

We focus on solving *linear homogeneous recurrence relations with constant coefficients.* The goal is to find an explicit formula (a *closed form*) for a_n that depends only on n and the initial conditions.

4.1 Method of Characteristic Polynomial

Consider a linear homogeneous recurrence relation of order k:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}.$$

We look for solutions of the form $a_n = r^n$, where $r \neq 0$ is a constant to be determined.

Substituting $a_n = r^n$ into the recurrence:

$$\begin{aligned} r^n &= c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k} \\ r^n - c_1 r^{n-1} - c_2 r^{n-2} - \cdots - c_k r^{n-k} &= 0 \\ r^{n-k}\left(r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k\right) &= 0. \end{aligned}$$

Since $r \neq 0$, we obtain the **characteristic equation**:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k = 0.$$

The roots of this polynomial are called the *characteristic roots.*

4.2 Case 1: Distinct Real Roots

Theorem. If the characteristic equation has k distinct real roots $r_1, r_2, \ldots, r_k$, then the general solution to the recurrence relation is:

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \cdots + \alpha_k r_k^n,$$

where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are constants determined by the initial conditions.

Solve the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$ with initial conditions $a_0 = 2$, $a_1 = 5$.

Solution. The recurrence can be rewritten as $a_n - 5a_{n-1} + 6a_{n-2} = 0$. Characteristic equation: $r^2 - 5r + 6 = 0$. Factor: $(r-2)(r-3) = 0$, so $r_1 = 2$, $r_2 = 3$ (distinct). General solution: $a_n = \alpha_1 \cdot 2^n + \alpha_2 \cdot 3^n$.

Use initial conditions:

$$\begin{aligned} n = 0 &: \alpha_1 + \alpha_2 = 2, \\ n = 1 &: 2\alpha_1 + 3\alpha_2 = 5. \end{aligned}$$

Solving: subtract twice the first equation from the second: $(2\alpha_1+3\alpha_2)-2(\alpha_1+\alpha_2)=5-4 \implies \alpha_2=1$. Then $\alpha_1=2-1=1$. Thus, $a_n=2^n+3^n$.

[Fibonacci Numbers] Solve the Fibonacci recurrence $F_n=F_{n-1}+F_{n-2}$ with $F_0=0$, $F_1=1$.

Solution. Rewrite: $F_n-F_{n-1}-F_{n-2}=0$. Characteristic equation: $r^2-r-1=0$. Roots:

$$r_{1,2}=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2}.$$

Let $\varphi=\frac{1+\sqrt{5}}{2}$ (the golden ratio) and $\psi=\frac{1-\sqrt{5}}{2}$. General solution: $F_n=\alpha_1\varphi^n+\alpha_2\psi^n$.

Using initial conditions:

$$\begin{aligned} n=0 &: \alpha_1+\alpha_2=0,\\ n=1 &: \alpha_1\varphi+\alpha_2\psi=1. \end{aligned}$$

From the first, $\alpha_2=-\alpha_1$. Substitute into the second:

$$\alpha_1\varphi-\alpha_1\psi=1 \implies \alpha_1(\varphi-\psi)=1.$$

Now, $\varphi-\psi=\frac{1+\sqrt{5}}{2}-\frac{1-\sqrt{5}}{2}=\sqrt{5}$. So $\alpha_1=\frac{1}{\sqrt{5}}$, $\alpha_2=-\frac{1}{\sqrt{5}}$.

Thus, **Binet's formula**:

$$F_n=\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n-\frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

4.3 Case 2: Repeated Real Roots

Theorem. If the characteristic equation has a root r of multiplicity m (i.e., $(x-r)^m$ is a factor), then the terms r^n, nr^n, n^2r^n, $\ldots$, $n^{m-1}r^n$ are all solutions. The part of the general solution corresponding to this root is:

$$(\alpha_1+\alpha_2n+\alpha_3n^2+\cdots+\alpha_mn^{m-1})r^n.$$

If there are several repeated roots, the general solution is the sum of such expressions for each distinct root.

Solve $a_n=6a_{n-1}-9a_{n-2}$ with $a_0=1$, $a_1=6$.

Solution. Rewrite: $a_n-6a_{n-1}+9a_{n-2}=0$. Characteristic equation: $r^2-6r+9=0$, or $(r-3)^2=0$. Double root $r=3$ (multiplicity 2). General solution: $a_n=\alpha_1\cdot 3^n+\alpha_2n\cdot 3^n$.

Using initial conditions:

$$\begin{aligned} n=0 &: \alpha_1=1,\\ n=1 &: 3\alpha_1+3\alpha_2=6 \implies 3(1)+3\alpha_2=6 \implies 3\alpha_2=3 \implies \alpha_2=1. \end{aligned}$$

Thus, $a_n=3^n+n\cdot 3^n=(n+1)\cdot 3^n$.

[Triple Root] Consider $a_n=3a_{n-1}-3a_{n-2}+a_{n-3}$. Characteristic equation: $r^3-3r^2+3r-1=0$, or $(r-1)^3=0$. Root $r=1$ with multiplicity 3. General solution: $a_n=\alpha_1\cdot 1^n+\alpha_2n\cdot 1^n+\alpha_3n^2\cdot 1^n=\alpha_1+\alpha_2n+\alpha_3n^2$.

4.4 Summary of the Method

Algorithm for Solving Linear Homogeneous Recurrence Relations with Constant Coefficients:

Step 1: Write the recurrence in standard form:

$$a_n-c_1a_{n-1}-c_2a_{n-2}-\cdots-c_ka_{n-k}=0.$$

Step 2: Form the **characteristic equation**:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k = 0.$$

Step 3: Find all roots $r_1, r_2, \ldots$ of the characteristic equation along with their multiplicities.

Step 4: Write the general solution:

- For each distinct root r of multiplicity m, include terms:

$$(\alpha_1 + \alpha_2 n + \cdots + \alpha_m n^{m-1}) r^n.$$

Step 5: Use the initial conditions to set up a system of linear equations and solve for the constants α_i.

Remark. The method described here applies specifically to linear homogeneous recurrences with constant coefficients. Non-homogeneous recurrences require finding a particular solution (similar to the method of undetermined coefficients for differential equations), and recurrences with variable coefficients often require different techniques (e.g., generating functions).

Applications. Recurrence relations and their solutions are fundamental in analyzing the time complexity of recursive algorithms (e.g., divide-and-conquer algorithms like MergeSort), modeling population growth, computing combinatorial quantities (e.g., Catalan numbers), and many other areas in mathematics and computer science.